

$\overset{\sigma}{\mathcal{M}} \subseteq \mathcal{P}(\overset{\sigma}{X})$
algebra
 $\overset{\sigma}{X} \in \mathcal{M}$
 $A \in \mathcal{M}$
 $\mathcal{M} \Rightarrow A^c \in \mathcal{M}$
 $\{A_n\}_{n=1}^\infty \subseteq \mathcal{M}$
 $\bigcup_{n=1}^\infty A_n \in \mathcal{M}$
 $(X, \mathcal{M})$
mea-
sure
able
space
 $\mu :$
 $\mathcal{M} \rightarrow [0, +\infty]$
mea-
sure
 $(X, \mathcal{M})$
 $\mu(\emptyset) = 0$
countable
ad-
di-
tiv-
ity
 $\{A_n\} \subseteq \mathcal{M}$

$$\mu\left(\bigsqcup_{n=1}^\infty A_n\right)=\sum_{n=1}^\infty \mu(A_n).$$

$(X, \mathcal{M}, \mu)$
mea-
sure
space
 $\overset{\sigma}{(X, \mathcal{M}, \mu)}$
 (Upward
 con-
 tin-
 uity)
 $A_1 \subseteq A_2 \subseteq \dots \subseteq \mathcal{M}$
 $\mu(\bigcup_n A_n) = \lim_{n \rightarrow \infty} \mu(A_n)$
 (Downward
 con-
 tin-
 uity)
 $A_1 \supseteq A_2 \supseteq \dots \supseteq \mathcal{M}$
 $\mu(A_1) < \infty$
 $\mu(\bigcap_n A_n) = \lim_{n \rightarrow \infty} \mu(A_n)$
(1)

$$\begin{aligned}
 &B_1 = A_1 \setminus B_n = A_n \setminus A_{n-1} \\
 &n \geq 2 \\
 &\{B_n\} \\
 &\bigcup_{k=1}^n B_k = A_n \\
 &\mu\left(\bigcup_{n=1}^\infty A_n\right)=\mu\left(\bigsqcup_{n=1}^\infty B_n\right)=\sum_{n=1}^\infty \mu(B_n)=\lim_{n \rightarrow \infty} \sum_{k=1}^n \mu(B_k)=\lim_{n \rightarrow \infty} \mu(A_n).
 \end{aligned}$$

(2)
 $C_n = A_1 \setminus A_n$
 $\mu(A_1) < \infty$
 $(X, \mathcal{M})$